

Coarse mismatch vs. factor-level diagnosis

$$\text{LEMON}(e, T) = \sum_i w_i m_i, \quad \sum_i w_i = 1$$

A. Polarity error

Triple: Aspirin - inhibits - COX-1

Text: Aspirin activates COX-1.

MINE-style

nodes ok; edge x; score ≈ 0.50

LEMON-Factor

factor	w x m	part
participants	.30 x 1	.30
inhibition relation	.30 x 0	.00
negative polarity	.20 x 0	.00
direction + evidence	.20 x 1	.20

LEMON score = 0.50

Diagnosis: polarity / inhibition contradicted

B. Relation-class loss

Triple: Alan Bean - birthPlace - Wheeler

Text: Alan Bean lived in Wheeler, Texas.

MINE-style

nodes ok; edge x; score ≈ 0.50

LEMON-Factor

factor	w x m	part
participants	.40 x 1	.40
biographical relation	.60 x 0	.00

LEMON score = 0.40

Diagnosis: birth relation missing